SUPPORTING ADULT GUIDANCE

A line is a set of solutions

Equations, Inequalities and Simultaneous Relationships · Lesson 5. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson makes the conceptual leap from a single-value solution to a **solution set**: a linear equation in two variables is satisfied by infinitely many (x, y) pairs, and those pairs are precisely the points of a straight line. It reuses the Dynamic Graph Lab to plot lines, test points, and read solutions off — groundwork for solving two relationships simultaneously in the lessons that follow.

Driving Question: *If x and y can both change, what does "solution" mean now?*

The mathematics, in plain terms

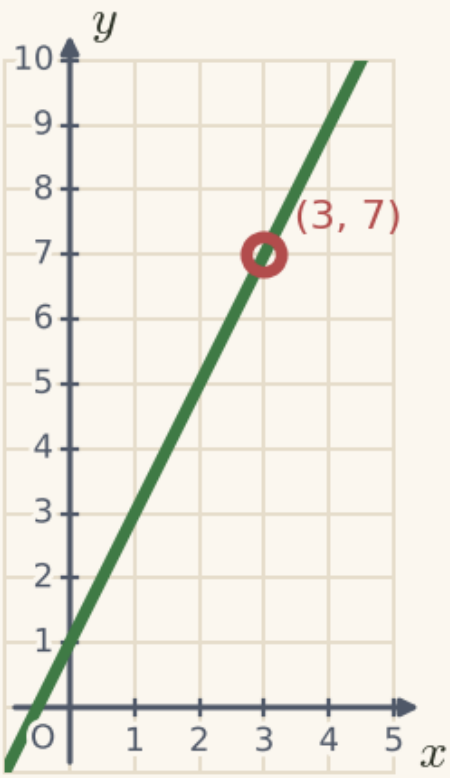
An equation such as y = mx + c (or x + y = k) constrains two variables but does not fix them; each choice of x determines a y, giving a pair, and the set of all such pairs is a straight line. Membership is decidable by substitution: (x■, y■) is a solution iff y■ = mx■ + c, i.e. iff the point lies on the line. This 'a line is the set of its solutions' equivalence is the bridge to reading a graphical intersection as a common solution.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Is this point a solution?

Is (3, 7) a solution of $y = 2x + 1$?

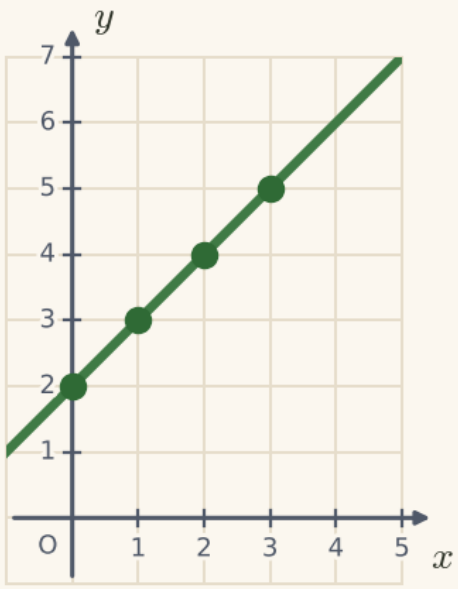


1. Substitute the point's x into the right-hand side.	$2 \times 3 + 1 = 7$
2. Compare with the point's y.	$7 = 7$ ✓
3. Both sides match, so the point is on the line.	(3, 7) is a solution

Yes — 2×3 + 1 = 7 equals the y-coordinate, so (3, 7) lies on the line and is a solution. (3, 5) would give 7 ≠ 5, so it is not.)

Example 2 — A table becomes the line

Plot solutions of $y = x + 2$ from a table.



1. Choose $x = 0, 1, 2, 3$ and work out $y = x + 2$.	$y = 2, 3, 4, 5$
2. Plot the pairs (0,2), (1,3), (2,4), (3,5).	four points
3. They fall on one line — the set of all solutions.	$y = x + 2$

The four pairs lie on one straight line; the table and the line are the same set of solutions, shown two ways.

From one answer to a whole set

The key shift is what 'solution' means: not a number to find, but a pair — and a whole line of them. Build the idea concretely: a table of pairs, plotted, becomes the line; then test points on and off it by substituting; then read a new solution off. Keep substitution as the honest test throughout — 'on the line' and 'makes the equation true' are the same statement.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Add pairs to $x + y = 6$ and watch them line up — many solutions, not one.	~9 min

Movement	Turn a table of (x, y) pairs into the line they form.	~10 min
Reflection	Test points on and off the line by substituting — the line is the solution set.	~12 min
Creativity	Read a solution off the line both ways ($x \to y$ and $y \to x$).	~8 min
Rest	A short pause on an answer widening into a whole set.	~2 min
Nutrition	Intellectual: the power of seeing an entire solution set at once.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking a two-variable equation has one answer.
Show two pairs that both work: 'Does (0, 6) satisfy $x + y = 6$? Does (1, 5)? So how many answers are there?'

Reading coordinates in the wrong order.
A point is (x, y): across first, then up. Ask 'which number is x, which is y?' before substituting.

Judging on/off the line by eye instead of substituting.
Make substitution the test: 'Put the point's x and y into the equation — do the two sides match?'

Assuming only whole-number pairs count.
The line is continuous: (2.5, 6) is on $y = 2x + 1$ just as much as (3, 7). The dots in a table are only samples of the whole line.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: coordinates, evaluating $y = mx + c$, and substituting a point into an equation. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says $y = 2x + 1$ 'has one answer', or checks a point by eye. The aim is a question that returns the reasoning to substitution.

- "Does (0, 1) work in $y = 2x + 1$? Does (1, 3)? So how many pairs are there?"
- "Which number in (3, 7) is x, and which is y?"

- “Substitute the point in — do the two sides come out equal? Then it is on the line.”
- “Pick any x you like — what y does the line pair with it? That is another solution.”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why does 'a line is a set of solutions' matter?

Because the next lessons solve two relationships at once: the point where two lines cross is the single pair that satisfies both — their common solution. That only makes sense once each line is understood as its own solution set.

Do we need the graph, or is substitution enough?

Substitution decides any single point, but the graph shows the whole set at once and lets you read solutions off — they are two views of the same thing, and seeing both is the point of the lesson.

Must solutions be whole numbers?

No — the line is continuous, so (2.5, 6) is as valid as (3, 7). A table only samples convenient whole-number pairs.

How is this different from Lesson 4?

Lesson 4 formed and solved for a single unknown. Here two unknowns vary together, so the answer is a set of pairs — a line — rather than one value.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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